MIS 7375 Homework 2: Pseudocode

Bayou City Urgent Care patient registration form | Sheldon McNeal | Version 2.1

Verbs used throughout: GET (read a value from the page), SET (give something a starting value), CALCULATE, DISPLAY, INDICATE (tell the user something is right or wrong), CHANGE. Structures: sequence, IF / THEN / ELSE, CASE, REPEAT FOR EACH.

Ingredients (inputs and working values)

```
INPUTS FROM THE PATIENT
    first name, middle initial, last name
    date of birth, preferred visit date
    social security number, gender
    email address, phone number
    address line 1, address line 2, city, state, zip code
    visited before, has insurance, flu shot this season
    pain level, medical history checkboxes, symptoms
    user ID, password, re-entered password

WORKING VALUES (not saved)
    today's date
    oldest allowed birth date, latest allowed visit date
    error message for each field
    all fields good (yes or no)
```

Getting the data off the form

```
REPEAT FOR EACH field name in the data dictionary
    GET the value the patient entered, using the field's name
    IF the field is a radio group THEN GET the choice that is selected
END REPEAT
REPEAT FOR EACH medical history checkbox
    IF checked THEN SET its value to Y ELSE SET its value to N
END REPEAT
Each check below receives these values and returns either nothing
(the field is good) or an error message. Only the main steps write
messages onto the page.
```

1. When the page opens

```
GET today's date from the browser
DISPLAY today's date in the banner, for example Thursday, October 15, 2026
CALCULATE the oldest allowed birth date as today minus 120 years
SET the date of birth limits from the oldest allowed birth date up to today
CALCULATE the latest allowed visit date as today plus 30 days
SET the visit date limits from today up to the latest allowed visit date
DISPLAY the pain level next to the slider, starting at 0 No pain
```

2. When the patient leaves a field

```
CHANGE the user ID to lowercase and trim a zip plus 4 to 5 digits
GET every value from the form by its field name
```

```
GET the name of the field the patient just left
CASE field name
    first name        : check first name
    middle initial    : check middle initial
    last name         : check last name
    date of birth     : check date of birth
    visit date        : check visit date
    ... one case for every field listed in section 3 ...
END CASE
IF the check found a problem THEN
    DISPLAY the error message in red beside the field
    INDICATE the field with a red border
ELSE
    CLEAR the error message and the red border
END IF
```

3. The check for each field

```
FIRST NAME
    IF blank THEN INDICATE first name is required
    ELSE IF it has anything other than letters, apostrophes or dashes
         or is longer than 30 characters
         THEN INDICATE letters, apostrophes and dashes only

MIDDLE INITIAL
    IF blank THEN it is fine, because it is optional
    ELSE IF it is not exactly one letter THEN INDICATE one letter only

LAST NAME
    IF blank THEN INDICATE last name is required
    ELSE IF it has anything other than letters, apostrophes, dashes
         or the numbers 2 through 5, or is longer than 30 characters
         THEN INDICATE the allowed characters

DATE OF BIRTH
    IF blank THEN INDICATE date of birth is required
    ELSE IF it is later than today THEN INDICATE it cannot be in the future
    ELSE IF it is earlier than the oldest allowed birth date
         THEN INDICATE it cannot be more than 120 years ago

PREFERRED VISIT DATE
    IF blank THEN INDICATE visit date is required
    ELSE IF it is earlier than today THEN INDICATE it cannot be in the past
    ELSE IF it is later than the latest allowed visit date
         THEN INDICATE it must be within the next 30 days

SOCIAL SECURITY NUMBER
    IF it is not 9 digits, with or without dashes
    THEN INDICATE the correct format

EMAIL ADDRESS
    IF it is not in the form name at domain dot ending
    THEN INDICATE the correct format
```

```
PHONE NUMBER
    IF it is not three digits, dash, three digits, dash, four digits
    THEN INDICATE the correct format

ADDRESS LINE 1
    IF blank or not 2 to 30 characters THEN INDICATE the problem

ADDRESS LINE 2
    IF blank THEN it is fine, because it is optional
    ELSE IF not 2 to 30 characters THEN INDICATE the problem

CITY
    IF blank or not 2 to 30 letters, spaces, periods or dashes
    THEN INDICATE the problem

STATE
    IF the patient left the starting "State" choice selected
    THEN INDICATE a state must be chosen

ZIP CODE
    IF it is not 5 digits or 5 digits, dash, 4 digits
    THEN INDICATE the correct format
    ELSE IF the patient entered the longer zip plus 4
         THEN CHANGE the zip code to only the first 5 digits
              AND DISPLAY the shorter zip code back in the box

VISITED BEFORE, HAS INSURANCE, FLU SHOT, GENDER
    IF no choice is selected in the group THEN INDICATE it was not answered

SYMPTOMS
    IF it contains a double quote mark THEN INDICATE double quotes are not allowed

USER ID
    CHANGE the user ID to all lowercase AND DISPLAY it back in the box
    IF it is not 5 to 30 characters THEN INDICATE the length rule
    ELSE IF it contains a space THEN INDICATE no spaces
    ELSE IF it starts with a number THEN INDICATE it cannot start with a number
    ELSE IF it has anything other than letters, numbers, underscore or dash
         THEN INDICATE the allowed characters
```

4. Password, checked while the patient types

```
EACH TIME the patient types a character in the password box
    CHECK whether the password
        is 8 to 30 characters long
        has at least one uppercase letter
        has at least one lowercase letter
        has at least one number
        has at least one special character
        has no double quotes and no spaces
        does not contain the user ID, first name or last name
    REPEAT FOR EACH rule in the list
```

```
        IF the rule is met THEN DISPLAY it in green with a check mark
        ELSE DISPLAY it in red with an X
    END REPEAT
    IF the re-enter password box is not empty
        THEN compare the two password boxes again

RE-ENTERED PASSWORD
    IF it does not match the password THEN INDICATE passwords do not match
```

5. Pain level slider, while the patient slides it

```
GET the slider number from 0 to 10
CASE slider number
    0         : SET the description to No pain
    1 to 3    : SET the description to Mild
    4 to 6    : SET the description to Moderate
    7 to 9    : SET the description to Severe
    10        : SET the description to Worst possible
END CASE
DISPLAY the number and the description next to the slider
```

6. REVIEW button

```
REPEAT FOR EACH field on the form
    check the field as in section 3
END REPEAT
DISPLAY the heading PLEASE REVIEW THIS INFORMATION
REPEAT FOR EACH item on the form
    DISPLAY one line with the item name, what the patient entered,
            and either pass or ERROR with the reason
END REPEAT
    For privacy, DISPLAY only the last 4 digits of the social security number
    For privacy, DISPLAY the password as stars only
    DISPLAY each medical history checkbox as Y or N
    DISPLAY anything the patient typed as plain text, never as page code
DISPLAY the review below the form and move the screen down to it
Do not send the form
```

7. SUBMIT button

```
The browser first checks the rules written into each field
(required, allowed characters, shortest and longest length, date limits)
IF any of those fail THEN the browser shows its own message and stops

SET all fields good to yes
REPEAT FOR EACH field on the form
    check the field as in section 3
    IF the field has a problem THEN SET all fields good to no
END REPEAT
IF all fields good is no THEN
    INDICATE "Please fix the fields marked in red before submitting"
    DISPLAY the review so every problem is listed in one place
```

```
    Do not send the form
ELSE
    send the form and DISPLAY the thank you page
END IF
```

8. START OVER button

```
CLEAR every field on the form
CLEAR every error message and red border
CLEAR the colors on the password rule list
CLEAR and hide the review area
DISPLAY the slider description as 0 No pain
```